

UNITED NATIONS ADAPTIVE SETTLEMENT INITIATIVE (UN-ASI)
DIRECTORATE OF EXTRAPLANETARY OPERATIONS



OBERON SYSTEMS DIVISION

FLEET-STANDARD PROCEDURES FOR POD DEPLOYMENT, HULL SUBDIVISION,
AND PLANETARY NON-PERMANENT SETTLEMENT FRAMEWORK

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This manual contains restricted settlement-architecture logic intended for Extraplanetary Settlement Vessels operating under UN-ASI Directive 2248-B. Unauthorized distribution is prohibited. All procedures herein must comply with planetary non-impact habitation protocols.

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SEE ALSO NOTES:

OSD-40253 –

OSD-97554 –

HULL SUBDIVISION STANDARDS

INELUTABLE POD ENVIRONMENTAL PROTOCOLS

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Introduction: Context + Scenario

In the mid-23rd century, Earth's environmental and geopolitical conditions reached a breaking point. Centuries of resource depletion, erratic climate patterns, and regional instability transformed long-habitable zones into unpredictable terrains. Coastal cities migrated inland; inland cities collapsed outward. Entire nations adopted rotational habitation cycles, moving populations seasonally between safe zones. Humanity remained technologically advanced but ecologically transient.

By 2248, the United Nations Adaptive Settlement Initiative (UN-ASI) commissioned a fleet of Extraplanetary Settlement Vessels enormous, semi-autonomous ships capable of landing on nearby moons and terrestrial-class planets for temporary occupation. These missions were not colonization projects. They were acts of planetary sampling, resource rotation, and short-term sanctuary, allowing human groups to survive without permanently scarring or exploiting their host worlds.

These settlement vessels were required to: identify safe landing locations on unknown terrain, deploy inflatable, airlocked habitation pods from their hulls, create semi-rigid structural frames from onboard scaffolding, adapt to local topography with minimal ground disturbance, and retract and depart fully intact when conditions required relocation.

Oberon was developed as the internal architectural logic system enabling these ships to evaluate terrain, orient themselves, subdivide their hulls, deploy pods, form scaffolding, and inflate temporary settlements all without permanent planetary impact.

This manual explains every part of that system. No step is trivial. Each component in Oberon exists because it ensures survival, adaptability, and non-destructive settlement on worlds that were never designed for us.

2. Preparation Requirements

Before Oberon can construct a functioning settlement deployment logic, several geometric and conceptual prerequisites must be completed. These ensure that all subsequent operations occur with precision and contextual awareness.

2.1 Planetary Terrain (Topography)

A settlement vessel cannot deploy without understanding where it is landing. The terrain provided to Oberon either as a mesh or surface is the fundamental dataset controlling every subsequent decision.

Why terrain matters:

The vessel cannot land on steep, uneven, or jagged surfaces. Pod deployment requires predictable, stable orientation. Scaffolding and inflated pods must react to true topographical constraints.

Important guidelines:

High-resolution meshes may be used but could burden computation. Simplifying excessively complex meshes ensures faster, clearer landing analysis. The topography defines the context of survival Oberon assumes nothing beyond what you give it.

2.2 Vessel Geometry (In-Flight + Landed Versions)

Because settlement vessels transform between flight and ground states, Oberon requires accurate representations of both.

In-Flight Vessel

Landing gear retracted. Clean geometry for alignment. Represents the orientation before atmospheric descent or orbital insertion.

Landed Vessel

Landing gear deployed. Resting configuration on planet. Defines the moment the ship becomes a temporary architectural anchor.

Why two versions?

Because in Oberon's logic: Landing location analysis happens before the vessel touches down. Pod deployment happens after the hull is grounded and stable.

Thus the system must understand both states.

The Simplified Hull

Both versions must contain a central hull object a minimal, clean geometry intended for subdivision. This is the "spine" from which pods deploy.

It must be simple because: UV subdivision requires logical parameterization. Complex, greebled geometry produces unpredictable patching. Pod deployment needs clean normals and consistent surface logic.

2.3 Bounding Box

The bounding box acts as a spatial and orientation reference for the vessel.

Why it matters:

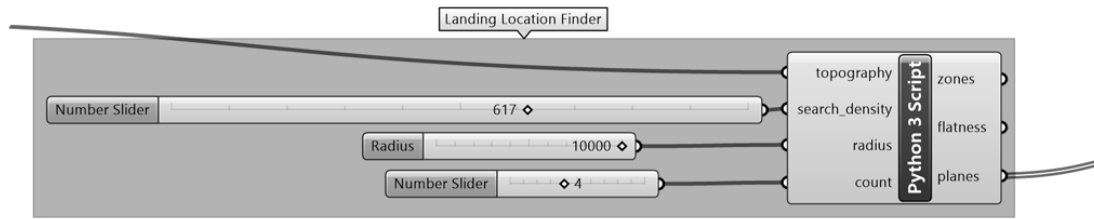
Ensures predictable vessel alignment to terrain planes. Assists in landing transformations, avoiding skewed or drifting placement. Helps determine collision-free descent orientation.

Oberon uses the bounding box as the vessel's "outer shell of certainty."

3. Component-by-Component Logic Explanation

Every Oberon component embodies part of a single chain of survival logic. Below is a deep, fully detailed explanation of each.

3.1 Landing Location Finder



Before any settlement deployment occurs, the vessel must determine where it should land not just where it can. This component analyzes the terrain to identify the safest and most viable landing zones.

Inputs (Expanded)

Topography

The planetary surface dataset. Represents the environmental unknown that the ship must interpret. All future architectural logic depends on this interpretation.

Search Density

Controls the number of sample points used during analysis. High density = meticulous caution (safer but slower). Low density = faster reconnaissance but may overlook micro-hazards. Essential for risk-adjusted landings on unpredictable worlds.

Radius

Defines the minimum stable surface footprint the vessel requires. Reflects the scale and weight distribution of the ship. Ensures that a "flat point" is not just a flat pixel but a flat region.

Count

Requests multiple viable landing zones. Useful because on alien terrain, one good zone may not be enough. Allows redundancy, simulation runs, or multi-vessel coordination.

Outputs (Expanded)

Zones

World-space coordinates of optimal landing spots. These are not arbitrary; they are computed safe zones.

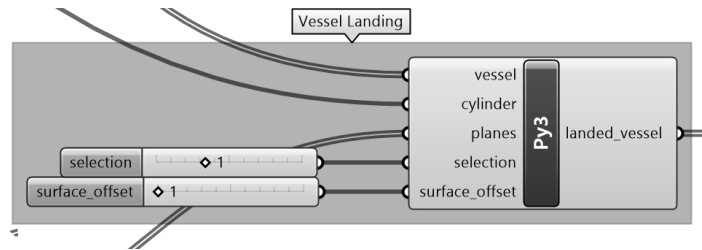
Flatness

A four-value diagnostic describing terrain suitability. May represent slope deviation, angular variance, and micro-topographical distortion. Allows designers to compare landings quantitatively.

Planes

Coordinate frames aligned to each landing zone. Necessary for aligning the ship in a physically meaningful orientation. Without these planes, the vessel would have no rotational logic when touching down.

3.2 Vessel Landing Component



Once viable landing zones are identified, the vessel must align and descend onto one. This component positions the vessel precisely.

Inputs (Expanded)

Vessel

The full vessel geometry in its in-flight orientation. Contains the simplified hull needed later.

Bounding Box

Ensures that the vessel's descent respects its true footprint. Avoids misalignment caused by uneven geometry.

Planes

The landing frames from the previous component. These are the "targets" or "hooks" for the landing.

Selection

Lets the user choose which landing zone to descend onto. Vital for multi-zone missions, convoy landings, or multi-scenario evaluations.

Surface Offset

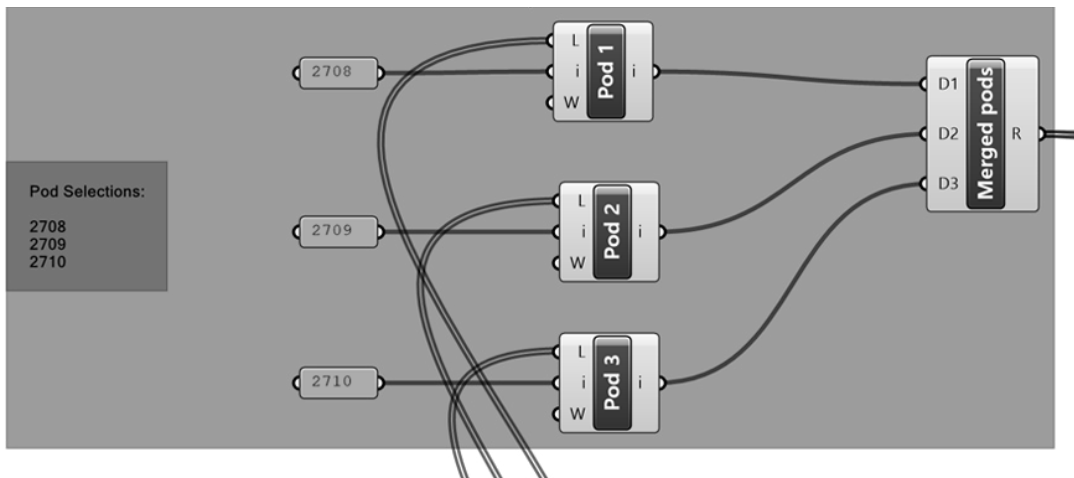
Provides visual clearance. Prevents interpenetration with noisy terrain meshes. Represents the real-world fact that vessels hover or cushion before full weight transfer.

Output

Landed Vessel

The vessel in its final, ground-truth orientation. This marks the moment where architecture begins.

3.3 Hull Extraction (Grasshopper Side Step)



Now that the vessel is grounded, Oberon needs the actual hull surface that will be used for pod deployment.

Why extraction is needed:

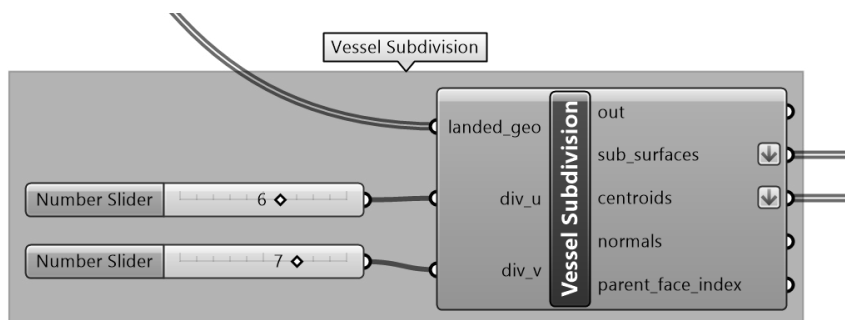
The in-flight hull may not match the landed orientation. The landed vessel may contain additional geometry not suitable for subdivision. Oberon must work with the "true" architectural surface in its final context.

Extraction Logic:

If one hull exists → direct connection. If multiple hulls exist → isolate via List Item → merge.

This ensures that pod logic acts upon the correct, stable, planetary-aware geometry.

3.4 Vessel Subdivision Component



This is where the vessel begins behaving like architecture.

Inputs (Expanded)

Landed Geometry

The simplified hull extracted from the grounded vessel. The surface that literally becomes the settlement wall.

U-Division & V-Division

Define the patch grid. Higher resolution = more deployment options. Lower resolution = coarser, more monumental pods.

Why subdivision exists:

Settlement pods must emerge from predictable, standard-sized modules. UV patches provide a consistent catalog of possible expansion nodes. Ensures geometric regularity essential for fabrication and inflation.

Outputs (Expanded)

Subdivided Surfaces

Each surface is a potential habitation node. Represents a conceptual "cell" in the vessel's architectural anatomy.

Centroids

Spatial anchors for pod generation. Provide positional logic for randomness, extrusion, and orientation.

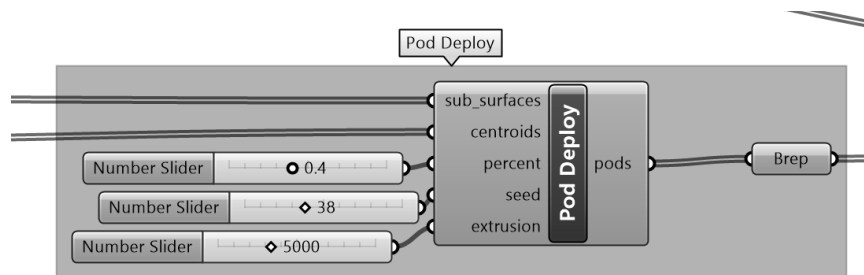
Normals

Define outward direction critical for pod deployment. Prevent pods from being sent into the hull interior.

Parent Face Index

Tracks which hull face generated which patch. Useful for filtering deployment by region or orientation.

3.5 Pod Deployment Component



Here the vessel begins expanding into settlement form.

Inputs (Expanded)

Subdivided Surfaces + Centroids

Foundation of pod placement logic. Must be flattened to unify patch indexing.

Percentage

Controls how much of the hull becomes living or working space. Reflects real mission constraints not all hull regions are serviceable.

Seed

Randomizes which patches become pods. Creates varied settlement layouts. Prevents predictable or vulnerable patterns.

Extrusion

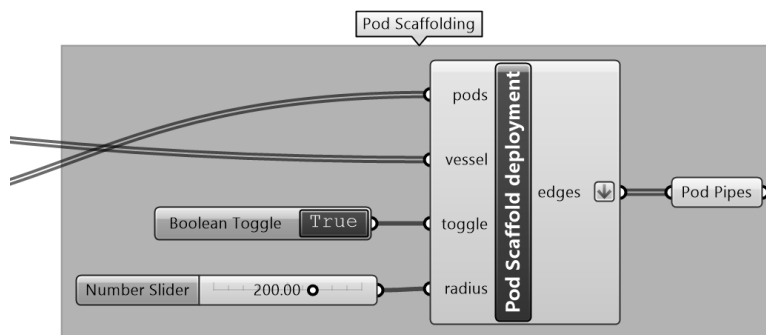
Determines pod depth. Represents required interior volume or pressure envelope.

Output

Extruded Pods

The first physical form of the settlement. Geometric representations of deployable, modular spaces.

3.6 Pod Scaffolding / Edge Component



Pods alone cannot survive alien conditions. They require structural support.

Why scaffolding exists:

Inflatable pods must be externally braced. Scaffolding maintains shape, protects against pressure variances, and locks pods into airlocked frames.

Inputs (Expanded)

Pods

The raw shells of the settlement.

Vessel Hull

Ensures scaffolding terminates cleanly at the hull boundary.

Toggle

Enables or disables frame generation.

Radius

Controls structural member thickness.

Output

Piped Scaffolding

A complete exoskeletal frame supporting pod integrity.

3.7 Pod Inflation Component (WIP)

In future iterations, pods will inflate into fully autonomous structures.

Planned Logic:

Vessel + scaffolding create a volumetric boundary. Inflatable pods expand until confined by these boundaries. Internal pressure parameter allows simulation-based deployment.

Planned Output:

Realistic, pressure-adjusted inflated pod volumes.

4. System Limitations

While Oberon provides a powerful logic framework for extraplanetary settlement, several conceptual and computational constraints remain:

4.1 Parametric Dependence

Oberon assumes consistent UV logic and clean topology. Highly irregular or greebled vessels may produce unpredictable subdivisions.

4.2 Mesh Dependency for Terrain

Noisy or overly dense terrain meshes can lead to false negatives in flatness analysis.

4.3 Idealized Physics

The current system assumes perfect extrusion, rigid scaffolding, and pressure-confined inflation without simulating: atmospheric variance, microgravity behavior, tectonic vibration, or thermal expansion.

4.4 Scaffolding Limitations

Scaffolding generation does not yet evaluate structural load paths or buckling scenarios.

4.5 Future Features Dependent on Pod Inflation System

The inflation system remains incomplete, limiting simulation accuracy for: airtight envelope behavior pressure differentials thermal membrane deformation

These limitations will be reduced in future versions.

5. Glossary

Adaptive Settlement Vessel

Mobile interplanetary craft designed for temporary habitation deployment.

Topography

Terrain surface defining planetary context.

UV Subdivision

Technique for dividing a surface into U and V parameter directions.

Pod Deployment

Process of extruding habitation units from subdivided hull surfaces.

Scaffolding

Structural frame supporting inflatable or rigid pod geometries.

Landing Plane

Computed orientation frame for safely positioning a vessel on terrain.

Search Density

Sampling resolution used to evaluate landing sites.

Flatness Metric

Numerical description of terrain suitability for landing.

Extrusion Distance

Depth or extent to which a pod expands outward from the hull.